

Ethanol Blending in Gasoline - Ecuador

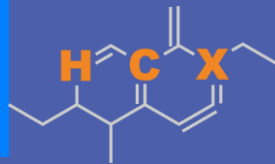
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**U.S. GRAINS &
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Ecuador

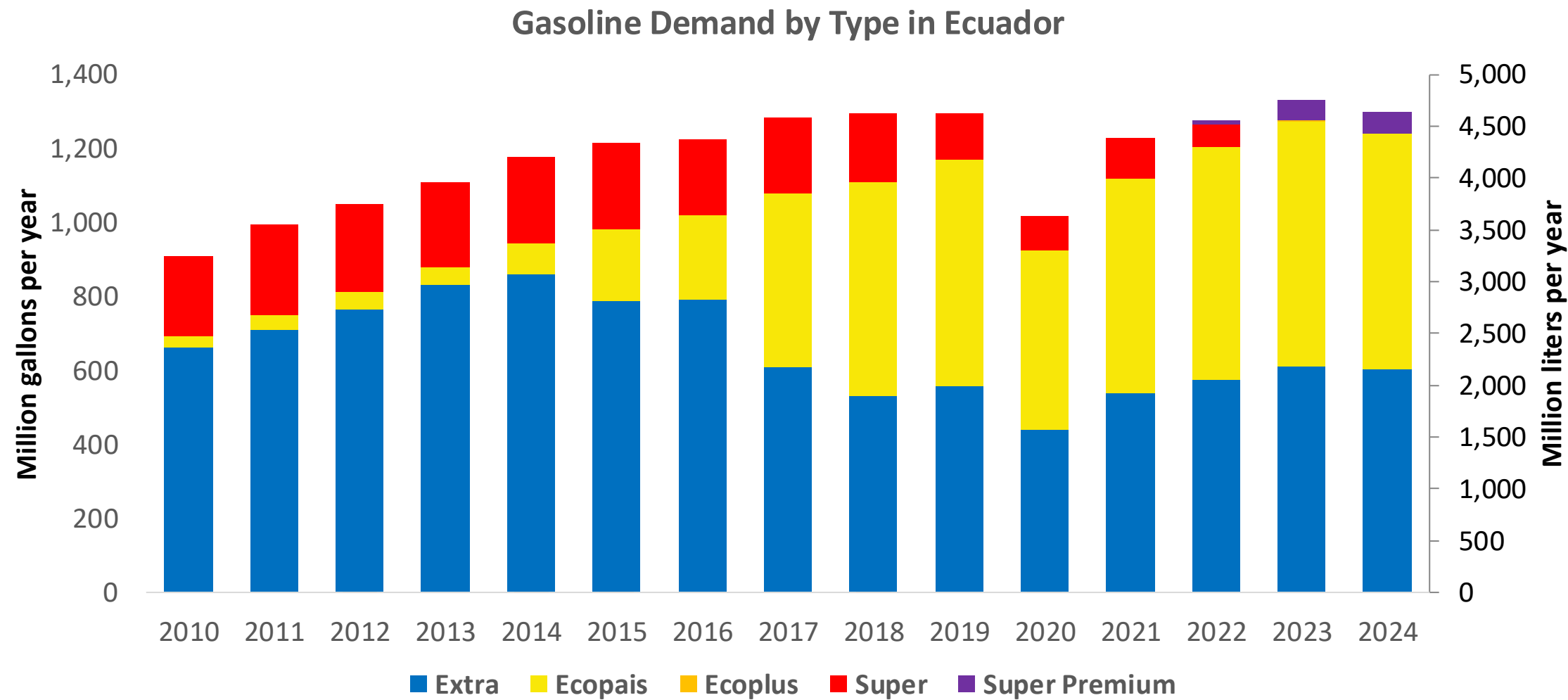


In 2024, gasoline consumption in Ecuador was 1,188 million gallons (4,498 million liters). A new plan was established to implement new grades of gasoline: four grades with 5 different types according to the current quality norm (NTE INEN 935:2021): Extra (RON 85), Ecopais E10 (RON 85), Ecoplus (RON 89), Súper (RON 92) and Super Premium (RON 95). Super Premium (RON 95) will gradually substitute Super (RON 92). Gasoline Ecoplus (RON 89) was recently introduced in the market. Ecuador has local production and imports naphtha (unfinished gasolines) to supply national demand.

There is no specific mandate for ethanol blending, however there is a permit to blend it with gasoline. Ethanol has been blending in gasoline since 2009. Today the usual blend is E10 for Ecoplus and Super. Ecopais gasoline is supplied in some provinces with an ethanol content between 3-4%. Gasoline Ethanol is produced, consumed and exported to neighboring countries for fuel and industrial use.

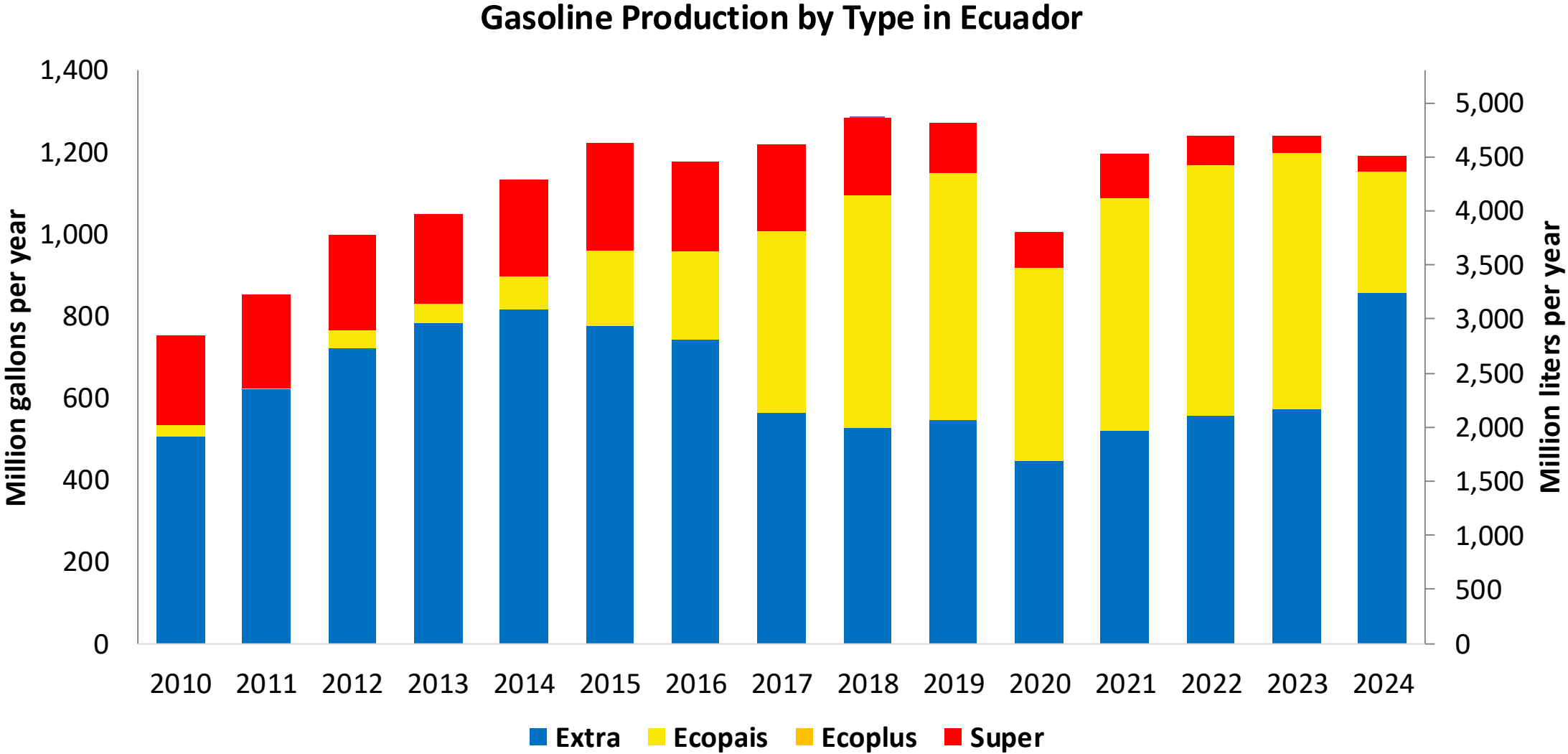
Source: Ministerio de Energía y Minas, EP PETROECUADOR, 2024

Gasoline Demand in Ecuador



Source: EP PETROEQUADOR, 2024

Gasoline Production in Ecuador

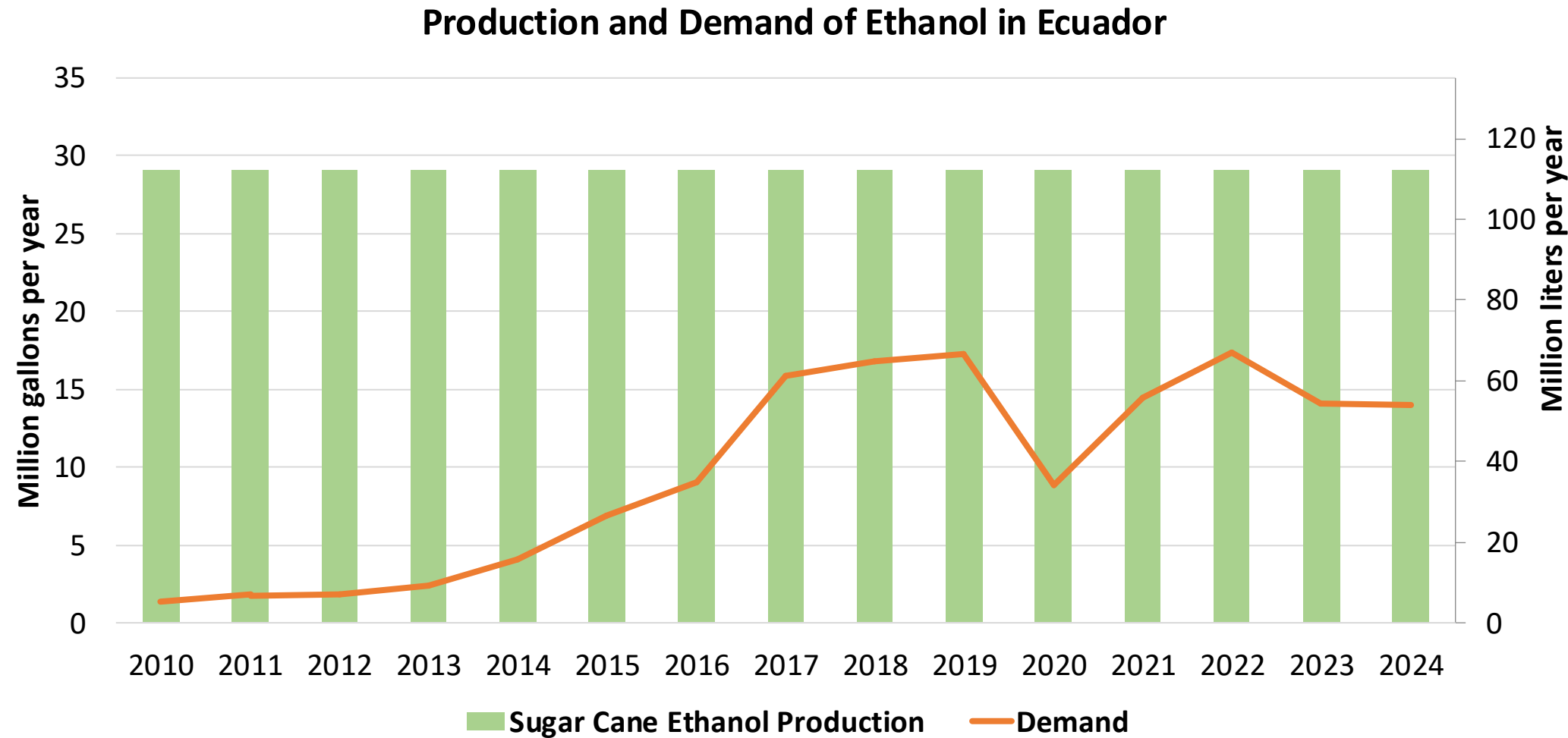


Source: EP PETROEQUADOR, 2024

The FARO90 logo is shown in a blue box, and a chemical structure is displayed to its right. The chemical structure features a central carbon atom (C) bonded to a hydrogen atom (H), a nitrogen atom (N), and a substituent X. The nitrogen atom is part of a ring system, and the substituent X is attached to a side chain.



Ethanol Balance in Ecuador



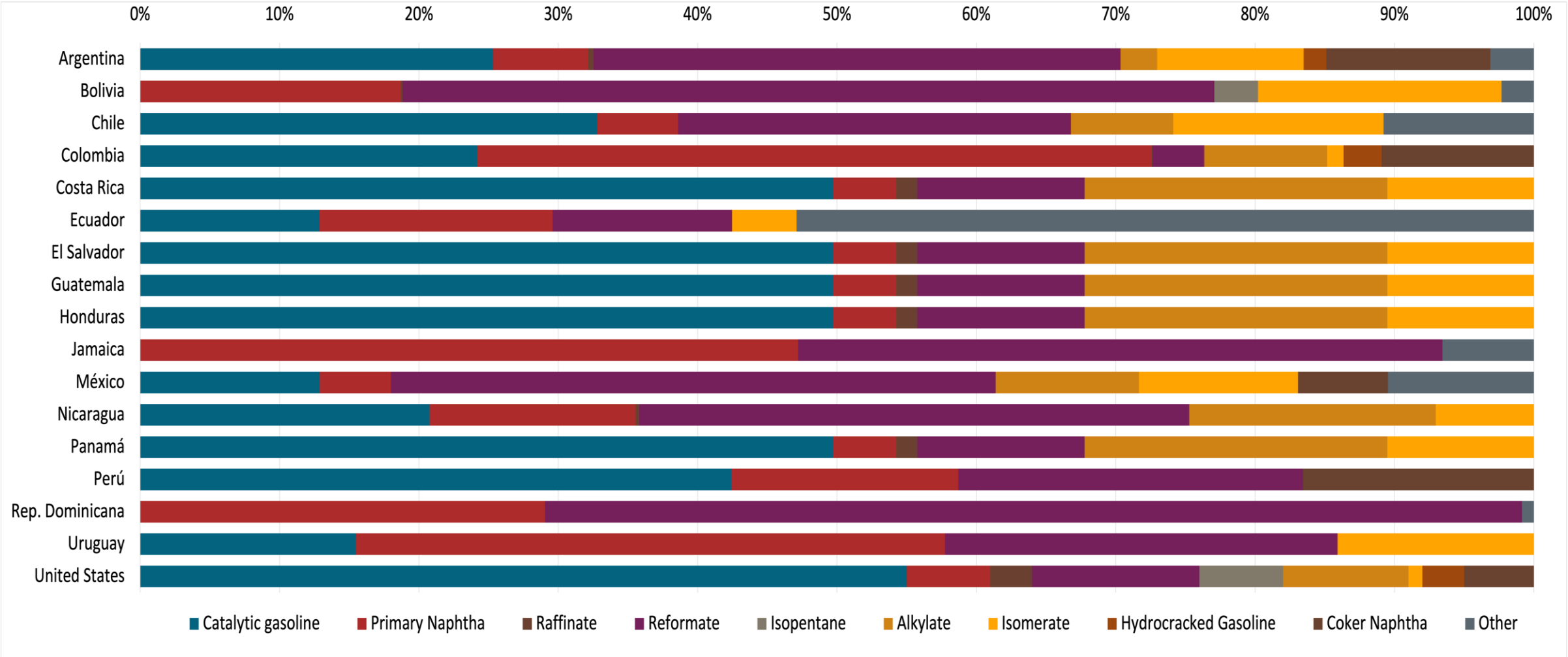
Source: EP PETROECUADOR, 2024; CEDRSSA, 2020

Gasoline Quality in Ecuador

Name	NTE INEN 935:2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	RON 85	RON 89	RON 92	RON 93	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 2 %v/v	<2 %v/v	<1,3 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 30 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 25 %v/v	< 25 %v/v	< 25 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	85	89	92	95	> 95	> 95	> 98	> 98
MON					> 85	> 88	> 85	> 88
AKI								
Sulfur Content	650 mg/kg	650 mg/kg	450 mg/kg	<300 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)					<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<60 kPa	<60 kPa	<60 kPa	<62 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



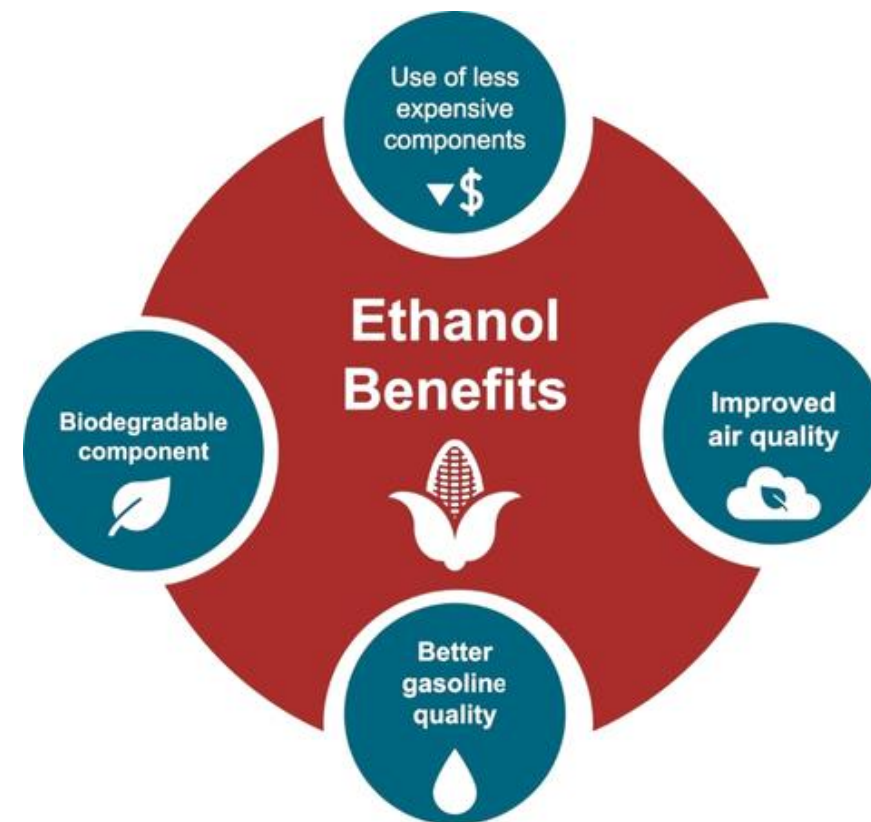
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

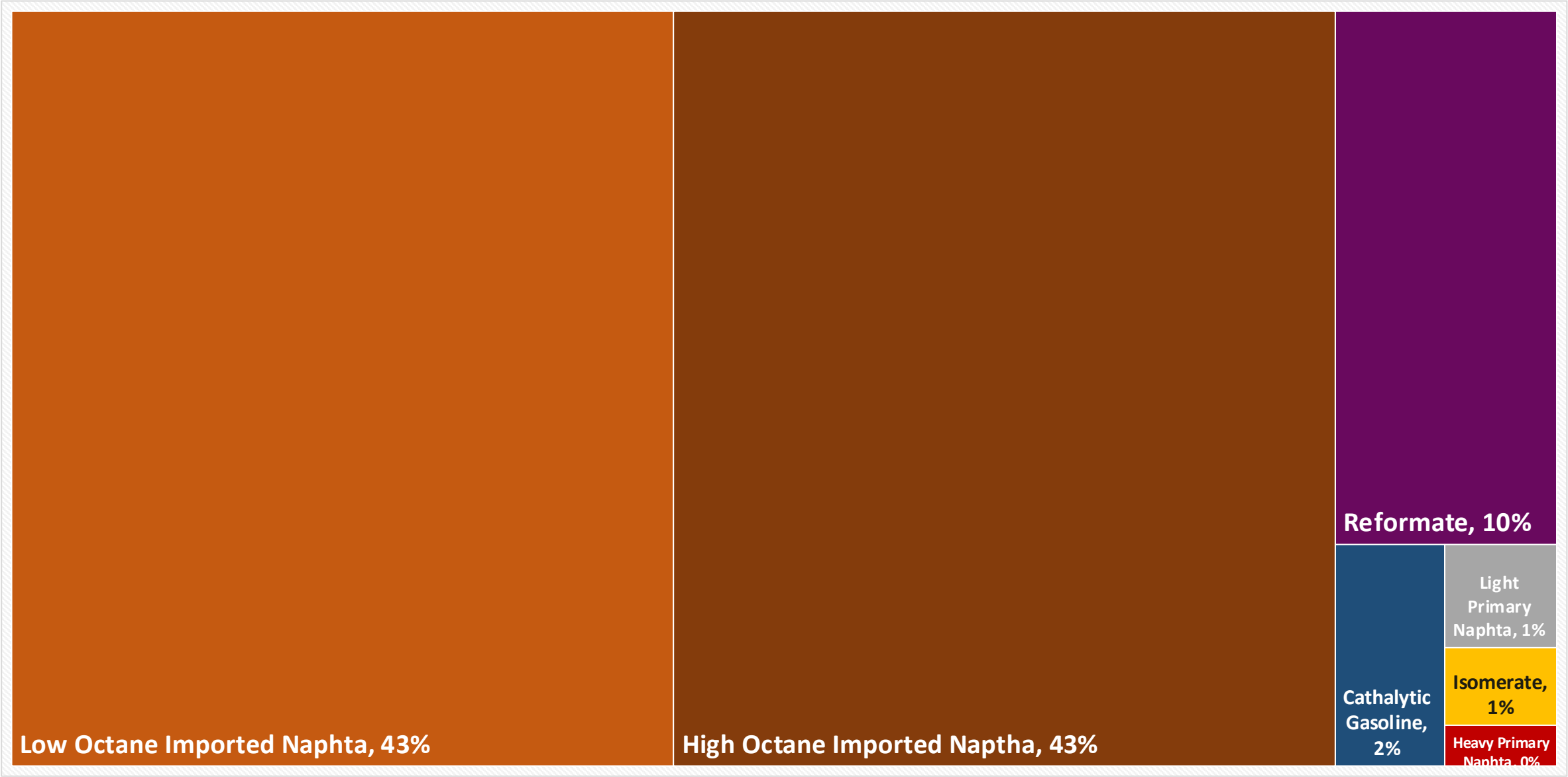
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

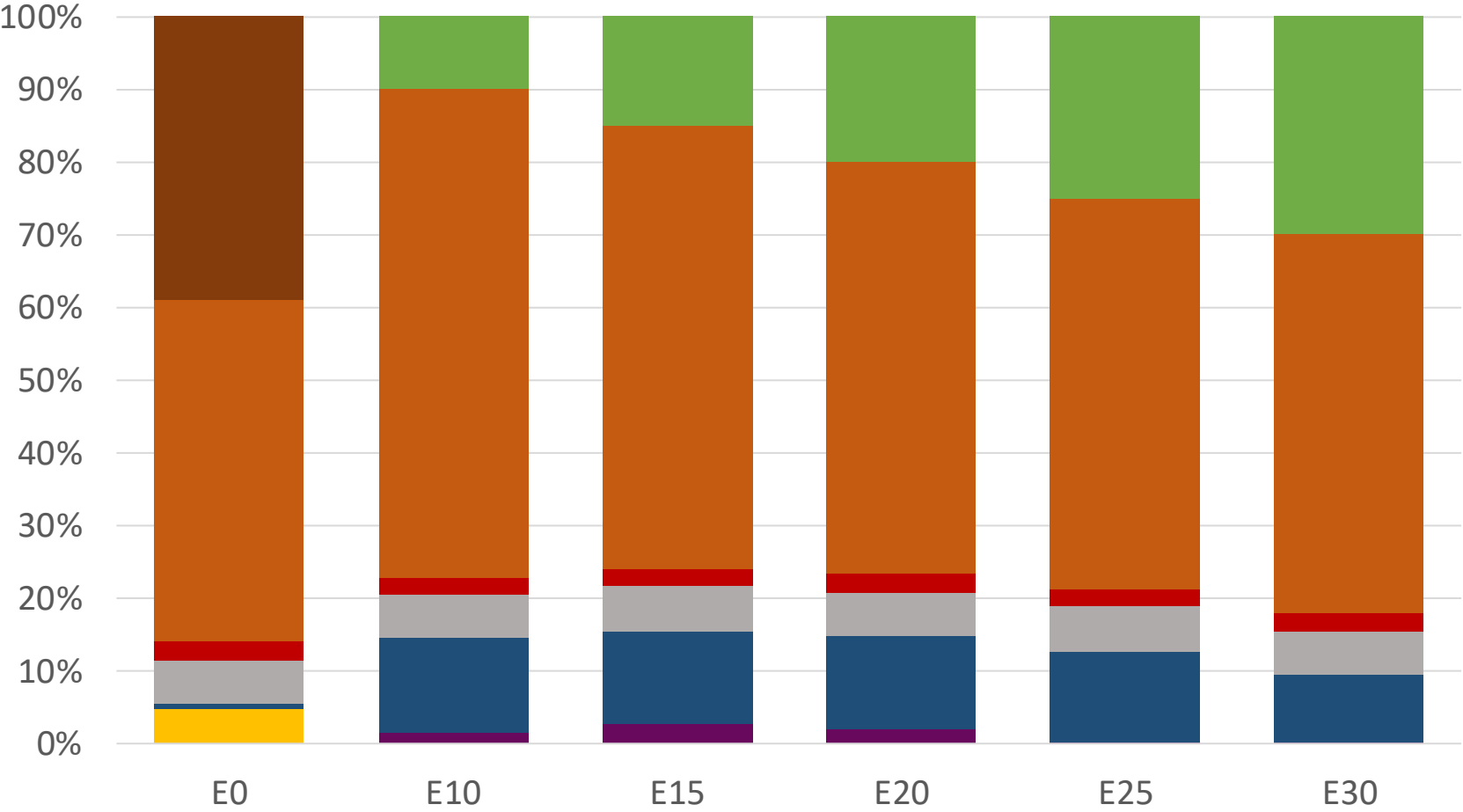
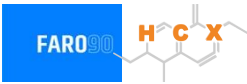
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

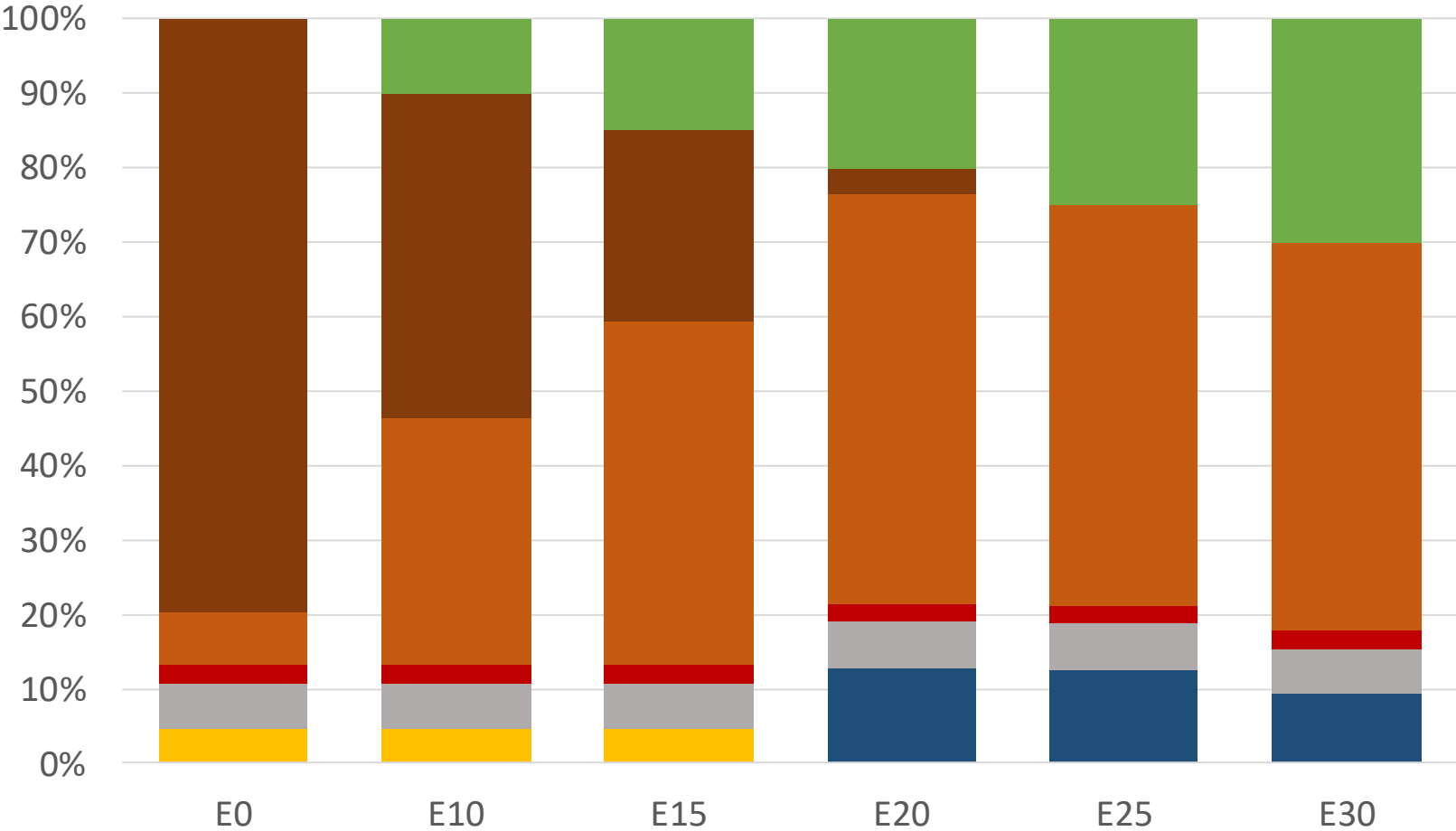


Ecuador –RON 85 Gasoline – Constant Octane



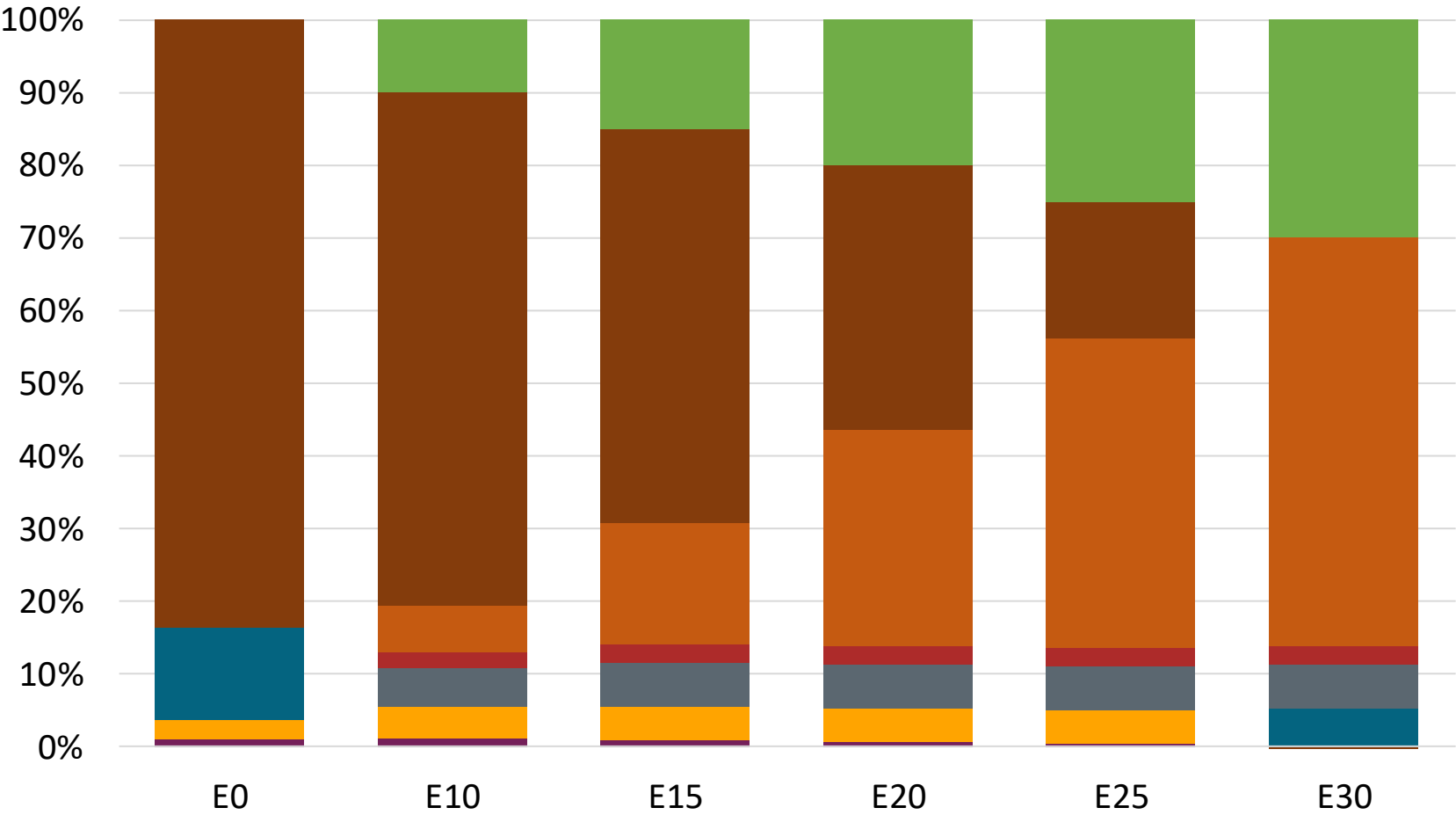
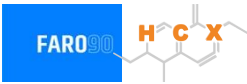
Octane (RON)	85.0	85.9	87.9	89.6	91.0	92.5
Price (US\$/gal)	\$ 2.386	\$ 2.321	\$ 2.310	\$ 2.288	\$ 2.258	\$ 2.234

Ecuador – RON 89 Gasoline– Constant Octane



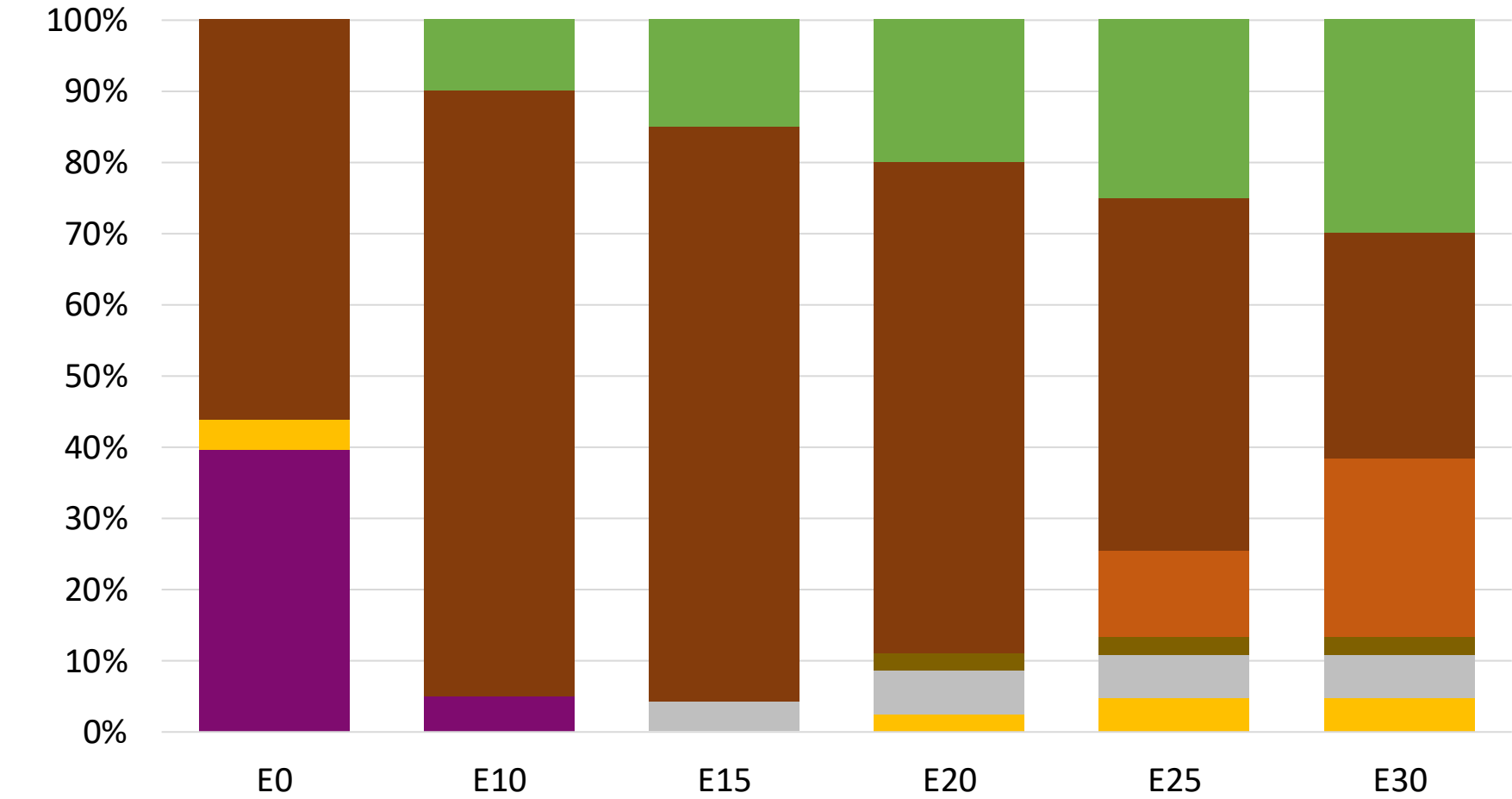
Octane (RON)	89.0	89.0	89.0	89.6	91.0	92.5
Price (US\$/gal)	\$ 2.453	\$ 2.357	\$ 2.310	\$ 2.282	\$ 2.258	\$ 2.234

Ecuador – RON 92 Gasoline – Constant Octane



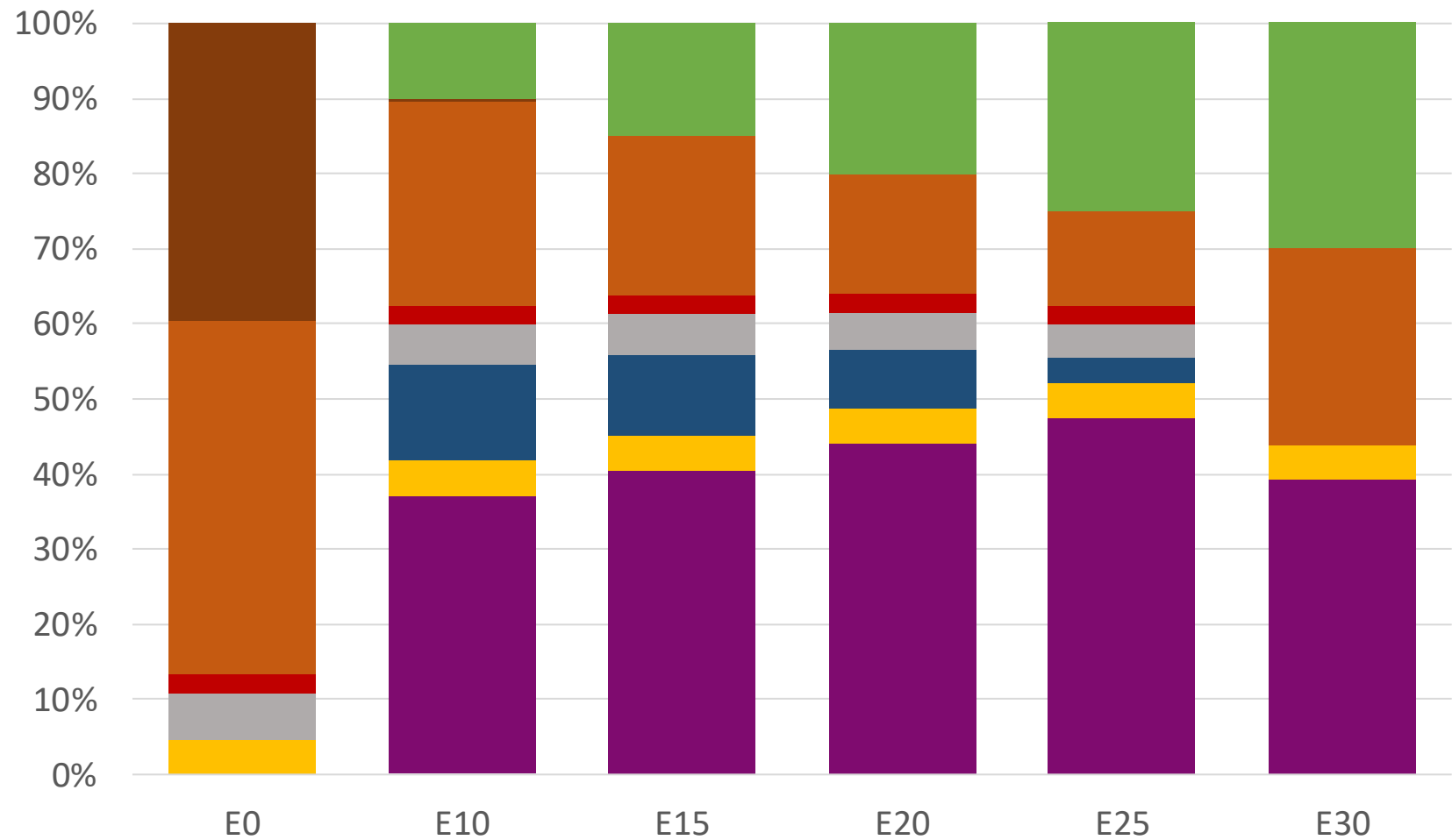
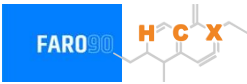
Octane (RON)	92.0	92.0	92.0	92.0	92.0	92.1
Price (US\$/gal)	\$ 2.584	\$ 2.419	\$ 2.362	\$ 2.314	\$ 2.266	\$ 2.225

Ecuador – RON 95 Gasoline – Constant Octane



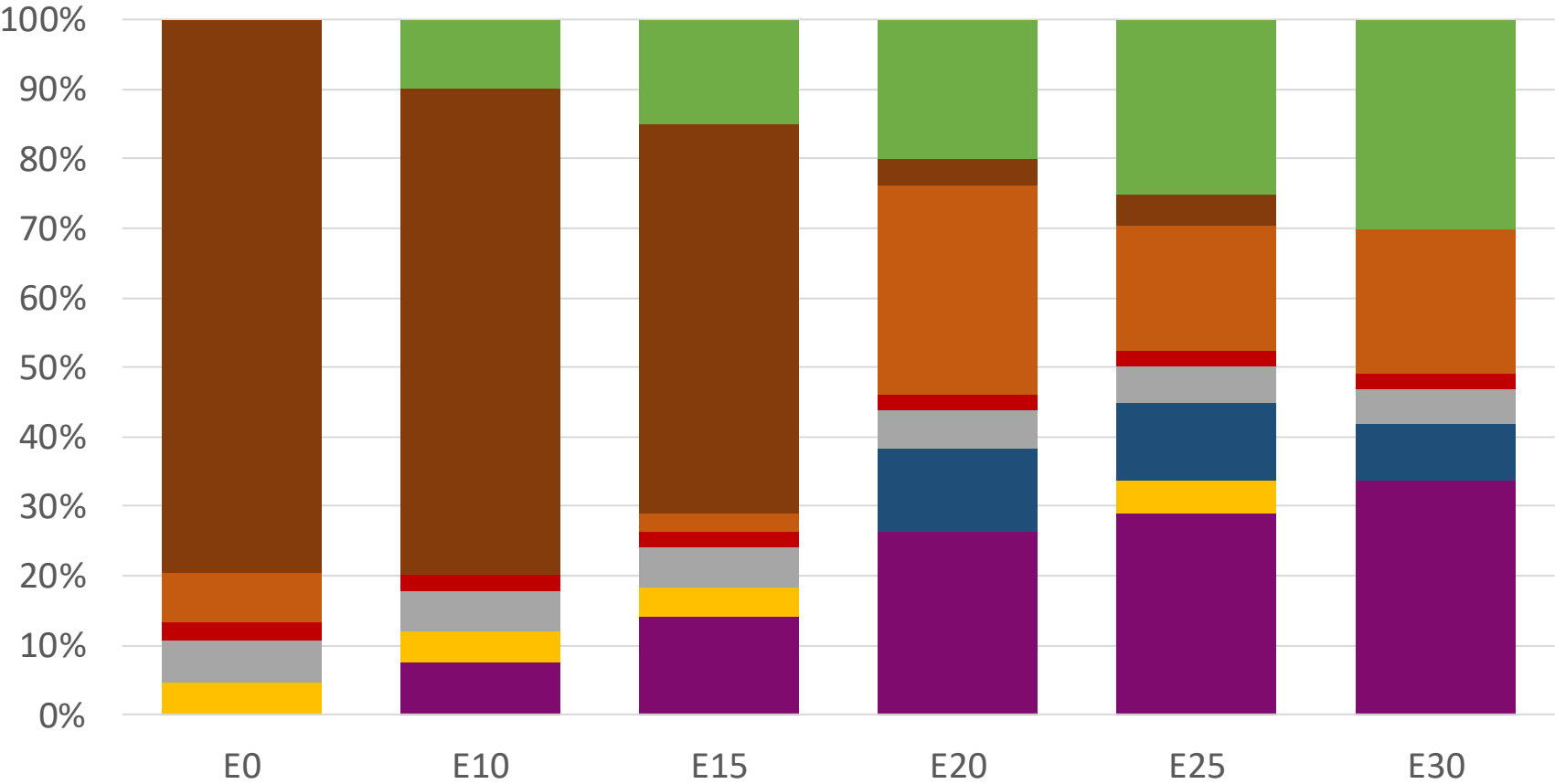
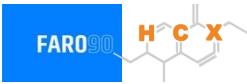
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.748	\$ 2.548	\$ 2.454	\$ 2.364	\$ 2.315	\$ 2.267

Ecuador – RON 85 Gasoline – Increased Octane



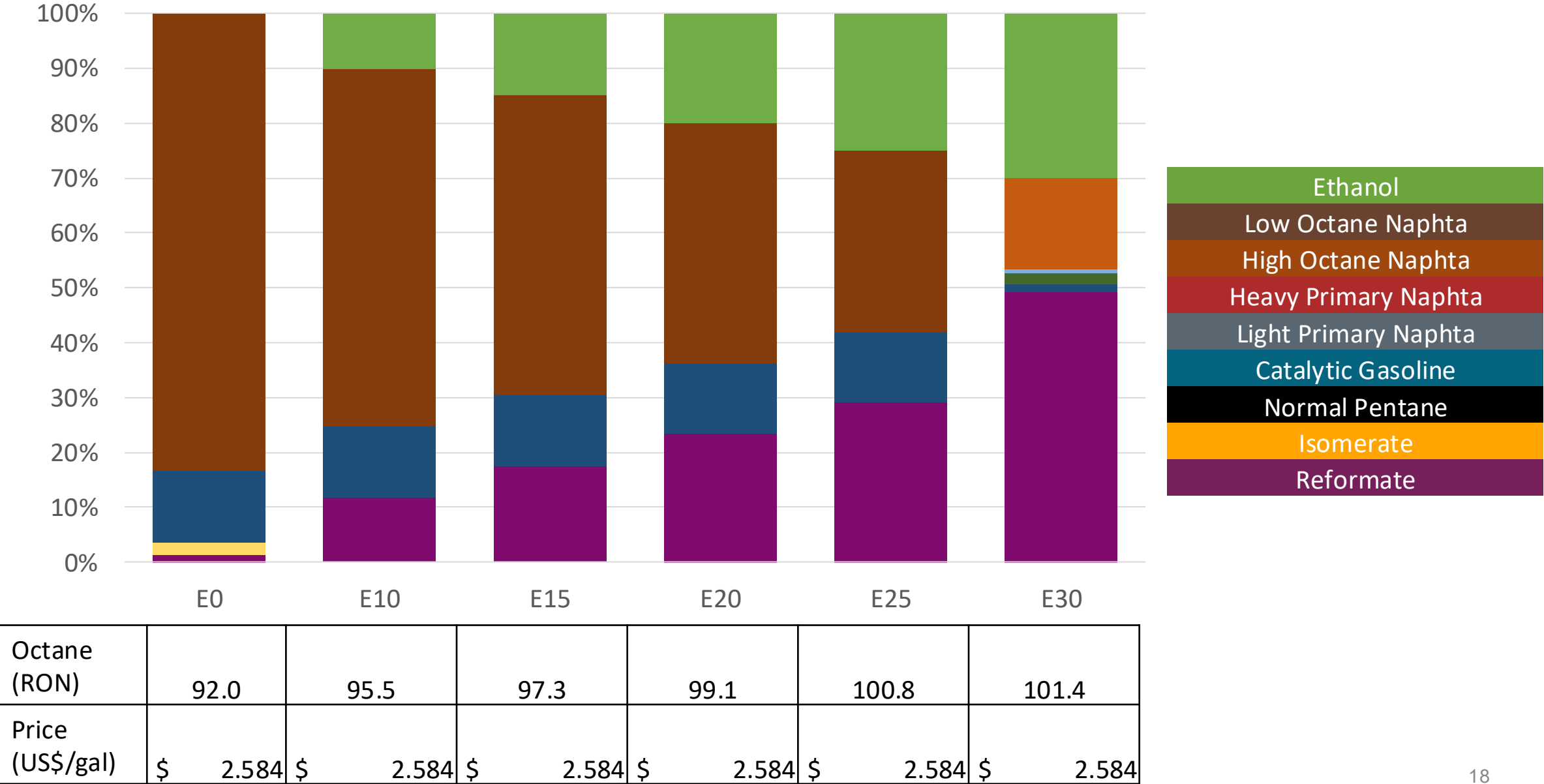
Octane (RON)	85.0	92.7	94.9	97.1	99.1	100.1
Price (US\$/gal)	\$ 2.386	\$ 2.386	\$ 2.386	\$ 2.386	\$ 2.386	\$ 2.386

Ecuador – Gasoline RON 89 – Increased Octane

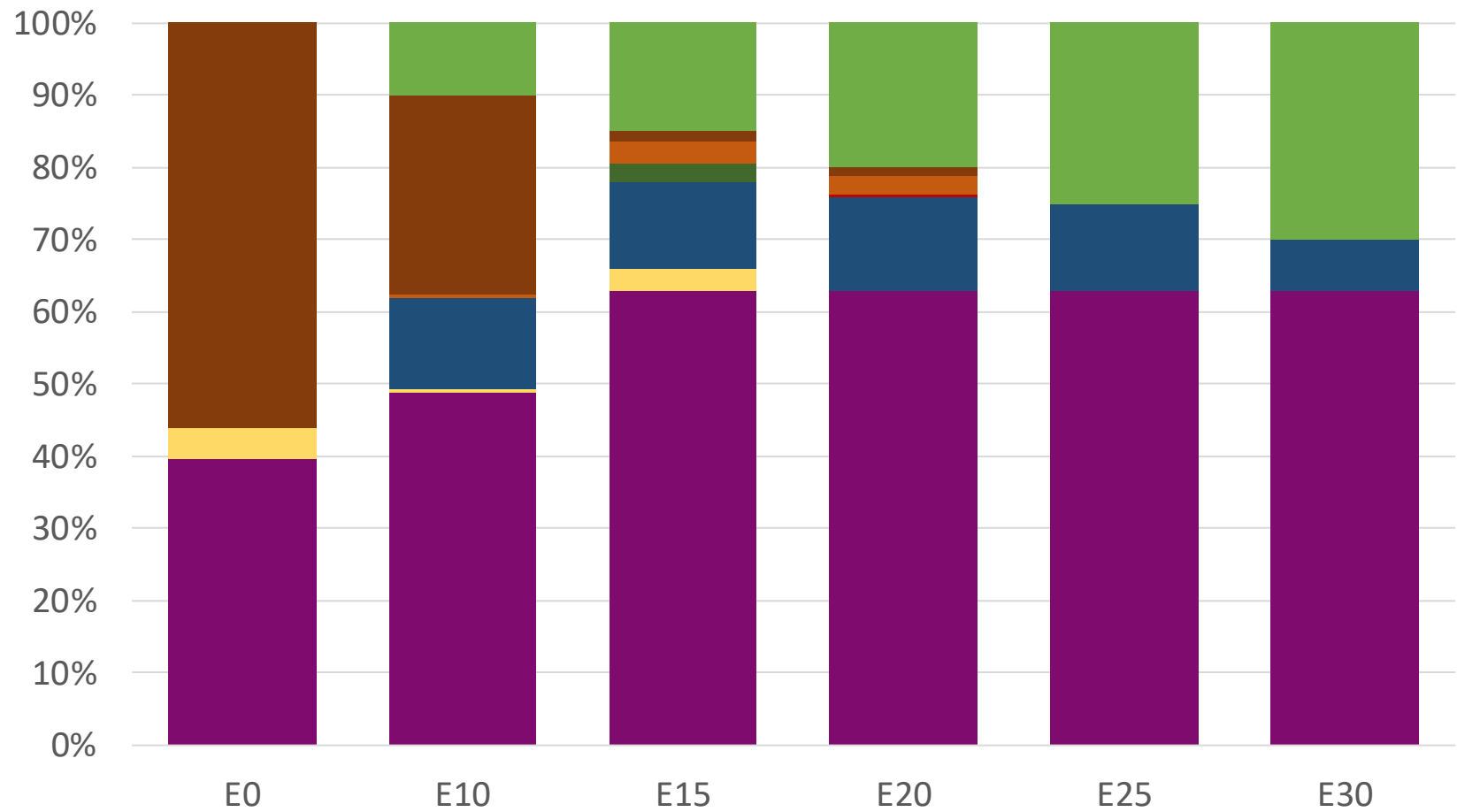
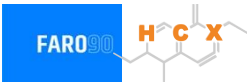


Octane (RON)	89.0	93.0	94.6	94.4	97.0	98.6
Price (US\$/gal)	\$ 2.453	\$ 2.453	\$ 2.453	\$ 2.453	\$ 2.453	\$ 2.453

Ecuador – RON 92 Gasoline – Increased Octane



Ecuador – RON 95 Gasoline – increased Octane



Octane (RON)	95.0	98.4	99.9	101.8	103.5	104.8
Price (US\$/gal)	\$ 2.748	\$ 2.748	\$ 2.748	\$ 2.748	\$ 2.748	\$ 2.748

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

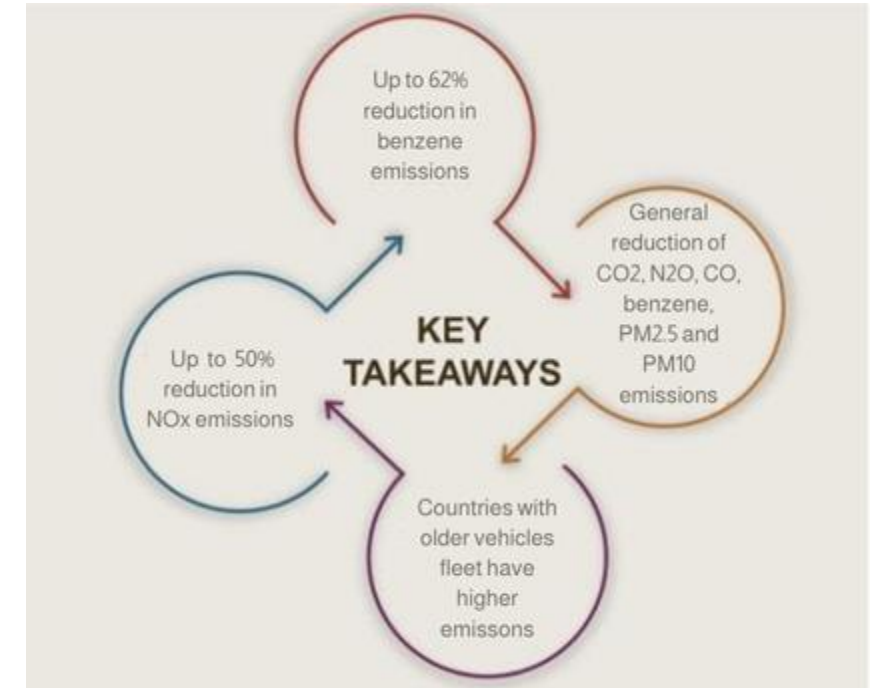
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

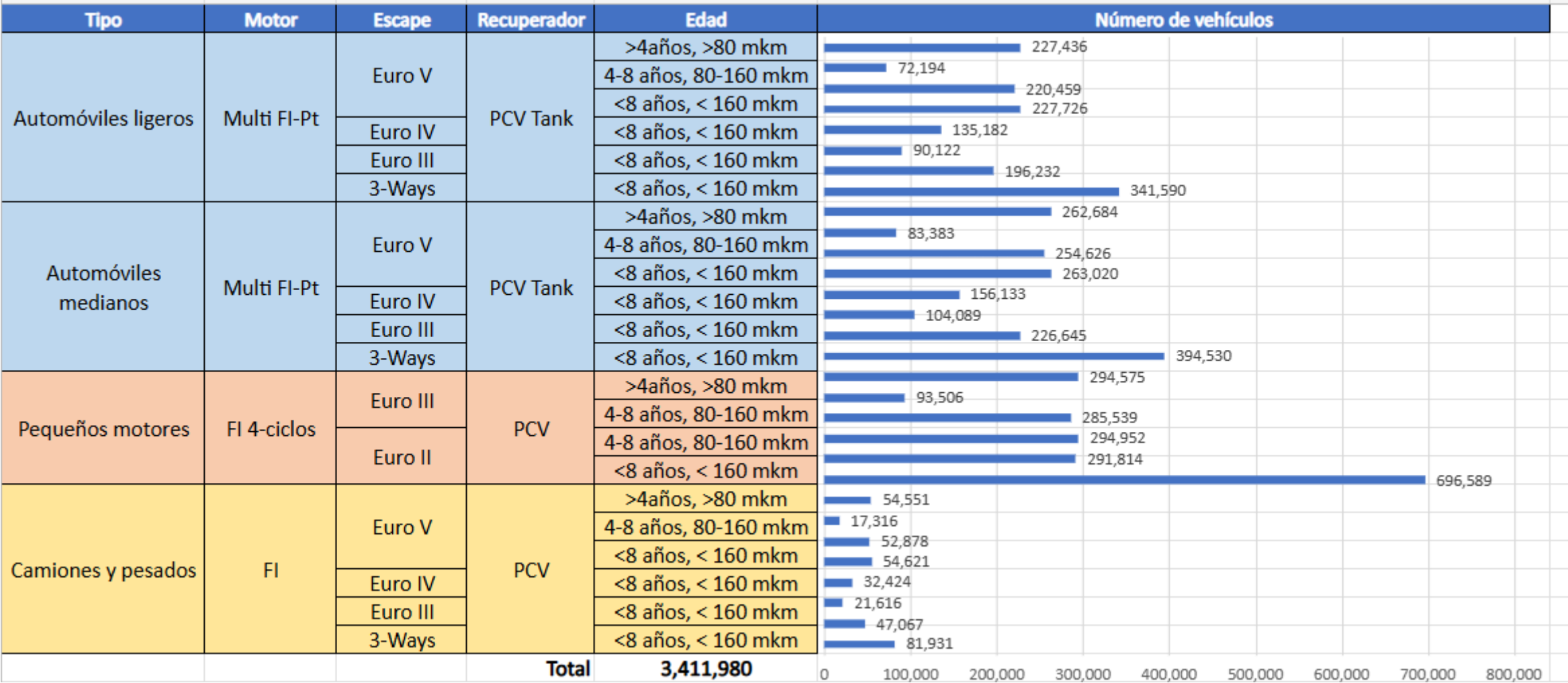
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Ecuador

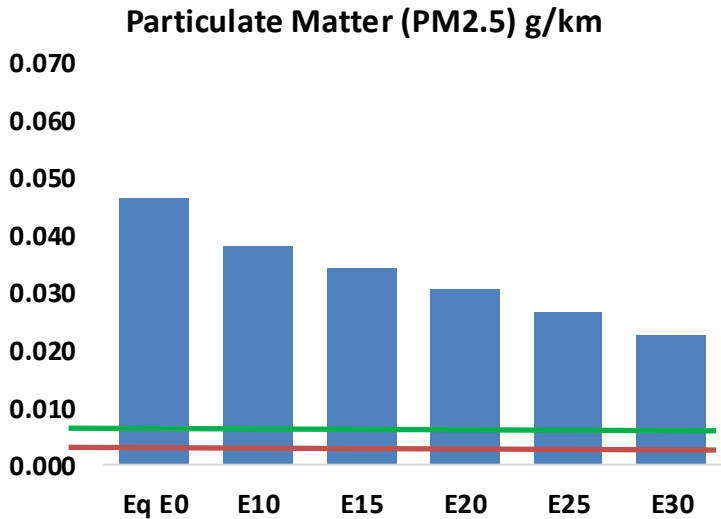
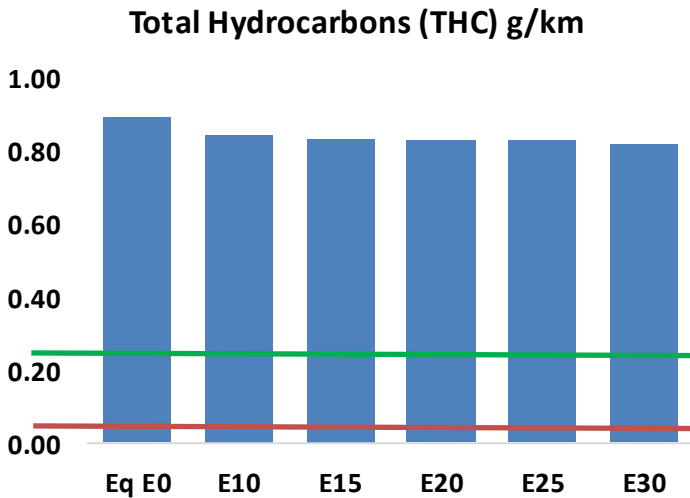
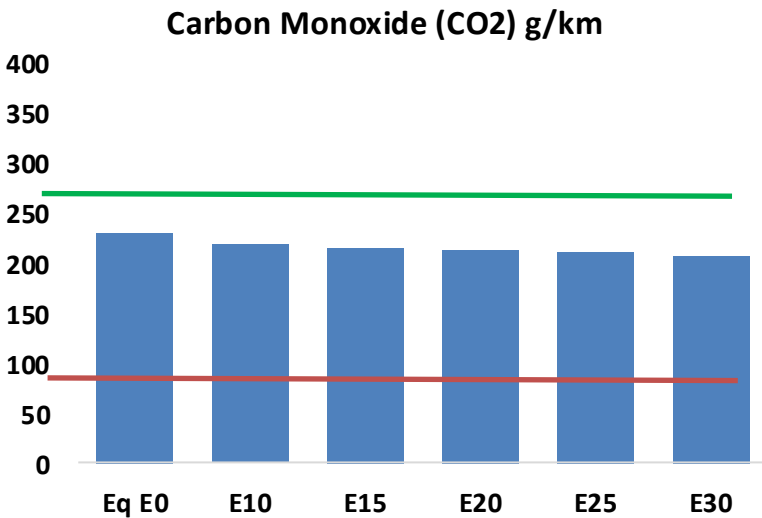
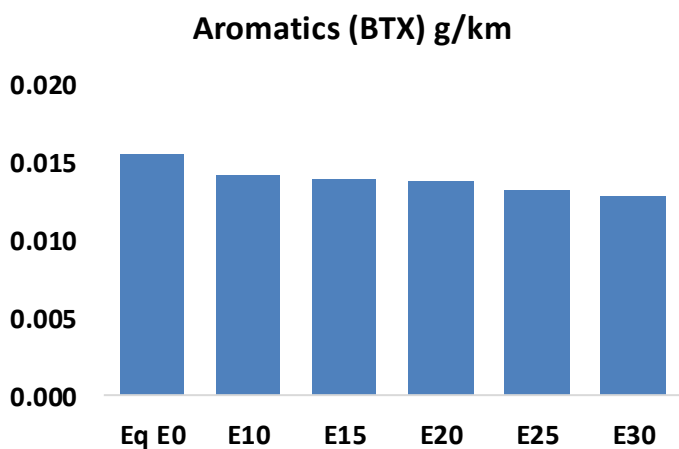
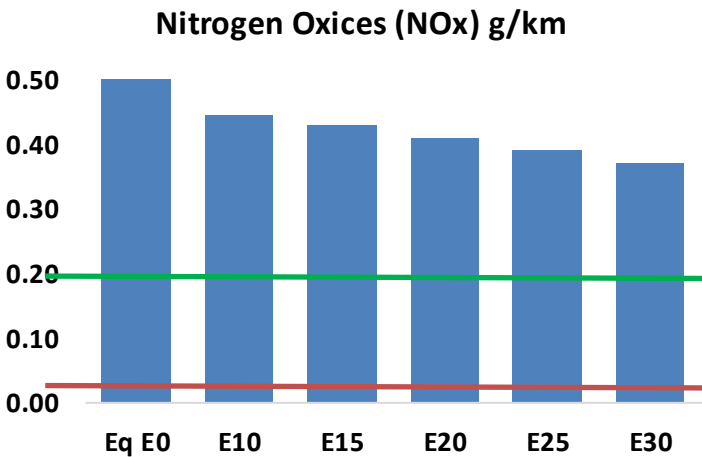
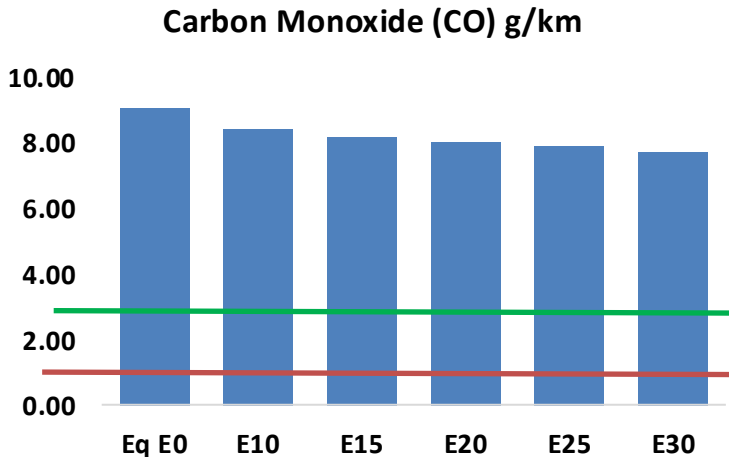
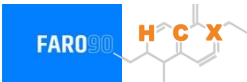


Vehicle Fleet: 3,411,980

Age: 13 years

Ecuador – Vehicular Emissions

United States
Euro 6



Ecuador – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	870,841	838,578	806,315	786,644	769,963	758,077	738,897	-7%	-12%
CO2	22,073,764	21,506,153	20,970,076	20,548,467	20,339,838	20,243,798	19,870,646	-5%	-8%
VOC	85,477	83,277	81,077	80,128	79,565	79,250	78,413	-5%	-7%
NOx	48,947	45,194	42,584	41,235	39,084	37,388	35,397	-13%	-20%
SOx	265	245	225	208	192	174	155	-15%	-28%
NH3	6,263	6,201	6,139	6,168	6,238	6,287	6,327	-2%	0%
N2O	824	819	816	820	837	846	856	-1%	2%
CH4	19,548	19,548	19,548	19,841	20,271	20,643	20,995	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	695	677	659	652	647	645	638	-5%	-7%
Acetaldehyde	1,415	1,624	2,383	3,628	4,943	5,649	6,674	68%	249%
Formaldehyde	5,271	5,124	5,974	7,014	7,348	8,129	8,850	13%	39%
Benzene	1,491	1,433	1,357	1,336	1,325	1,267	1,226	-9%	-11%
PM2.5	1,483	1,348	1,213	1,094	978	854	724	-18%	-34%
PM10	2,943	2,675	2,408	2,172	1,942	1,695	1,437	-18%	-34%

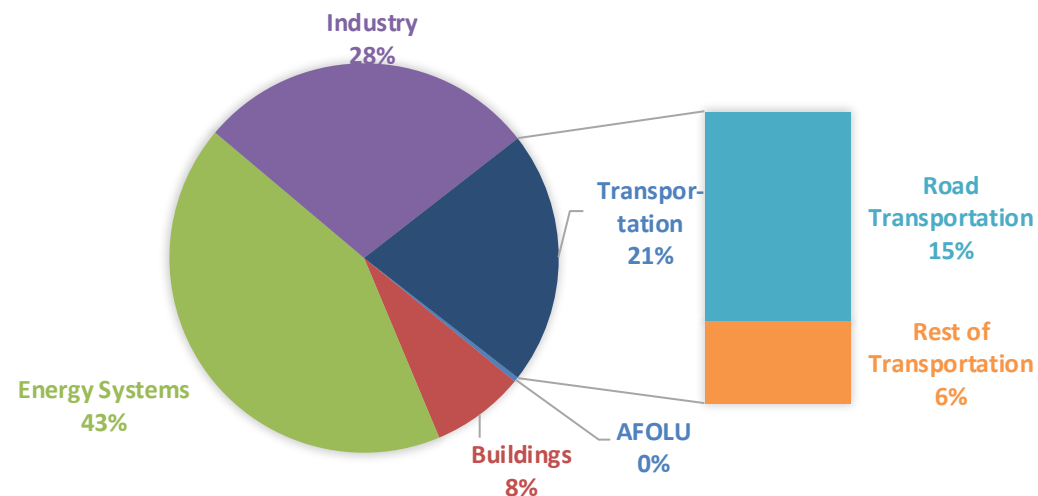
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

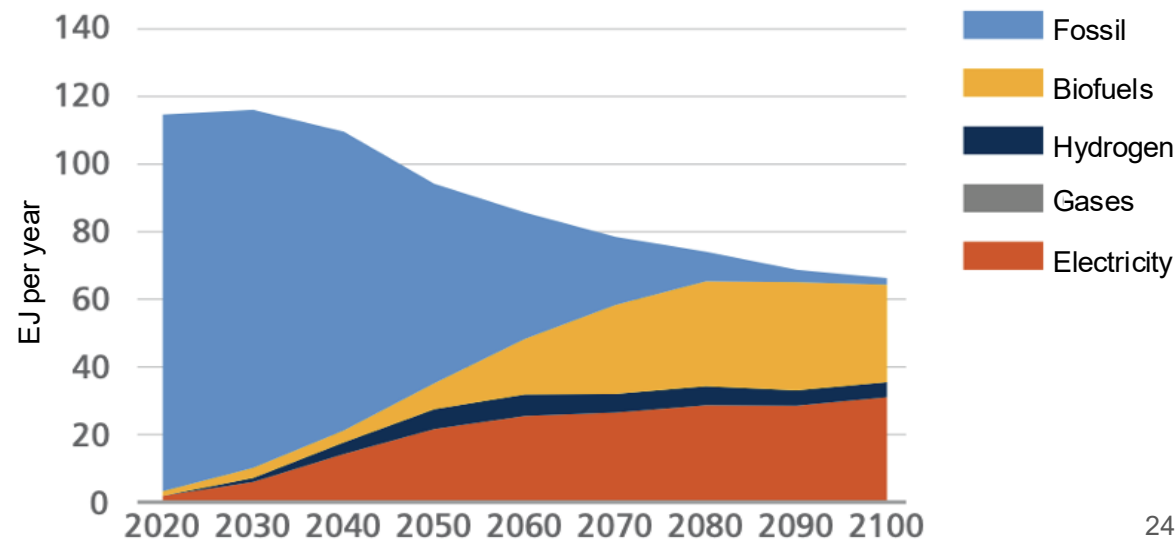
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

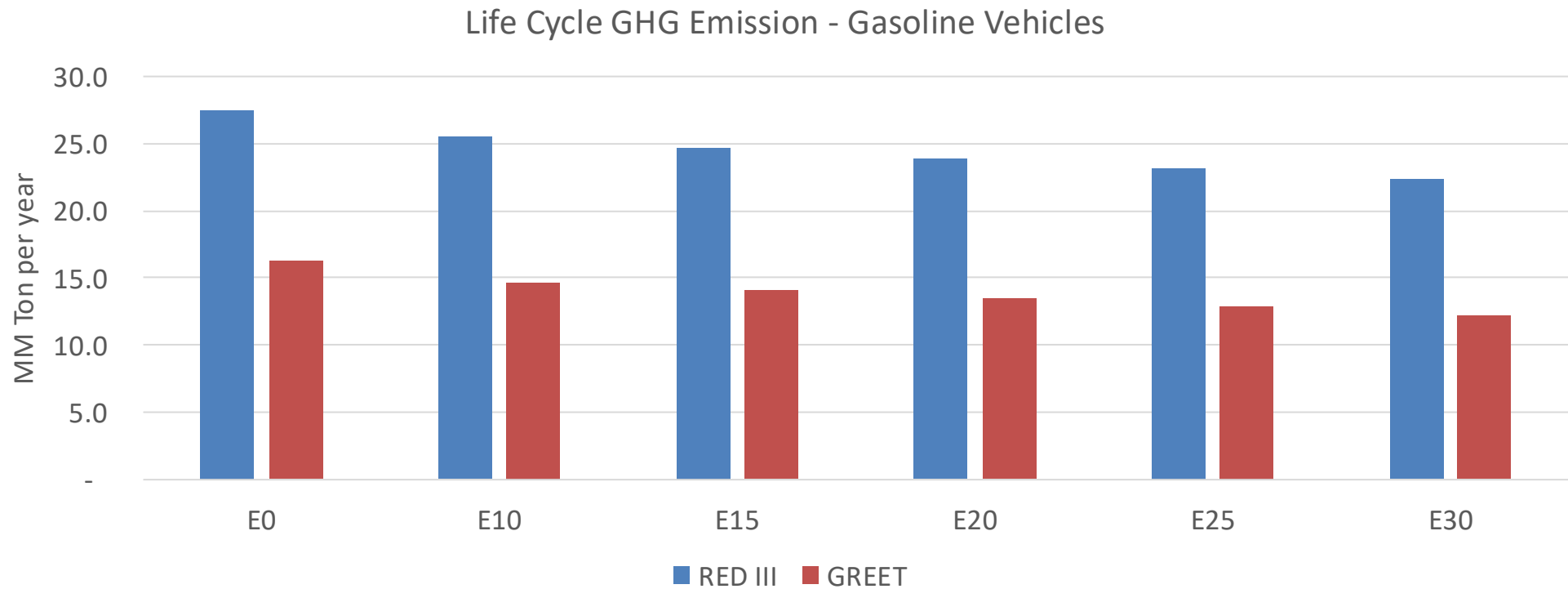
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Ecuador – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	101.4
2035 Target (MMT/year)	71.0
Delta	30.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.1%	13.7%	17.5%	20.7%	24.8%
RED II/III	0.0%	7.5%	10.4%	13.2%	15.8%	18.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.4%	7.3%	9.3%	11.1%	13.3%
RED II/III	0.0%	6.8%	9.4%	12.0%	14.3%	17.0%